

Technology Stack

Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

A Technology Stack refers to the combination of programming languages, frameworks, tools, databases, and platforms used to develop and run an application. It includes both front-end (client-side) and back-end (server-side) technologies, along with supporting tools like cloud services, APIs, and version control systems. A well-defined technology stack ensures that the system is scalable, efficient, maintainable, and aligned with the project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript (React.js)	Provides a responsive and interactive user interface with reusable components and real-time data visualization.

2	Backend / Server-Side	Python (Flask)	Lightweight framework suitable for ML integration, fast API development, and easy handling of prediction models.
3	Database / Data Storage	PostgreSQL	Reliable relational database with strong data integrity, suitable for storing structured weather and prediction data.
4	Cloud / Hosting / Deployment	AWS (EC2 / S3)	Scalable and cost-effective cloud platform for hosting applications and storing datasets securely.
5	Version Control & CI/CD	GitHub	Enables version control, team collaboration, and integration with CI/CD pipelines for automated deployment.
6	Third-Party APIs / Other Tools	OpenWeather API, Scikit-learn, XGBoost	OpenWeather API provides real-time weather data; Scikit-learn and XGBoost are used for building and deploying the flood prediction model.